

WAZUH Home Lab - SIEM and File Integrity Monitoring

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In this home lab I set up Wazuh, a free and open-source SIEM and XDR platform, to monitor a Windows machine from a centrally managed Ubuntu server. The goal was to get hands-on experience with how security events and logs are collected, forwarded, and visualized in a real security operations environment.

The lab walks through installing and configuring the Wazuh Manager on an Ubuntu VM, deploying the Wazuh Agent on a Windows host, connecting the two over a local network, and demonstrating File Integrity Monitoring (FIM), where any file created or deleted inside a monitored folder triggers a real-time alert on the dashboard.

Key Skills:

- Log analysis and aggregation
- File Integrity Monitoring (FIM)
- Intrusion detection
- Vulnerability detection
- Real-time alerting and dashboards

Tools Used

- Wazuh - open-source SIEM/XDR platform (manager, agent, dashboard)
- Ubuntu Server 20.04+ - hosts the Wazuh Manager inside VirtualBox
- VirtualBox - hypervisor used to run the Ubuntu VM
- Windows - host machine running the Wazuh Agent
- ossec.conf - Wazuh agent configuration file for enabling FIM

The key points were:

- Installed and configured the Wazuh Manager on an Ubuntu VirtualBox VM
- Accessed and explored the Wazuh web dashboard
- Installed and registered a Wazuh Agent on a Windows host
- Configured File Integrity Monitoring for a specific directory
- Verified real-time alert forwarding from the agent to the manager dashboard

1. Lab Architecture

Component	Host	Role
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Wazuh Manager	Ubuntu (VirtualBox VM)	Collects, analyzes, and stores data from agents. Hosts the web dashboard.
Wazuh Agent	Windows (host machine)	Sends logs, events, and FIM alerts to the Wazuh Manager.

1.1 Same Subnet

WINDOWS

ipconfig

```
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
```

UBUNTU

ifconfig

```
mtran@Ubuntuvm:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.68 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fe87:7509 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:87:75:09 txqueuelen 1000 (Ethernet)
    RX packets 375 bytes 27235 (27.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 81 bytes 10259 (10.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. Installing the Wazuh Manager (Ubuntu)

I Ran all of the following commands on the Ubuntu VM terminal.

2.1 Add the Wazuh GPG Key

This verifies the authenticity of Wazuh packages before installing them:

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o
/usr/share/keyrings/wazuh-archive-keyring.gpg
```

```
mtran@Ubuntuvm:~$ curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gp
g --dearmor -o /usr/share/keyrings/wazuh-archive-keyring.gpg
```

2.2 Download and Run the Installation Script

This single command downloads and runs the all-in-one installer, setting up the manager, indexer, dashboard, and Filebeat automatically:

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash ./wazuh-install.sh -a -i
```

The installation took around 7 minutes. When it finished, the terminal displayed the dashboard URL, username, and auto-generated password. saved all information.

```
mtran@Ubuntuvm:~$ curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash ./wazuh-install.sh -a -i
27/03/2026 18:54:51 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
  User: admin
 Password: Fm8D.sbQPZXiNtXhEe?aYatKMow+cXD7
```

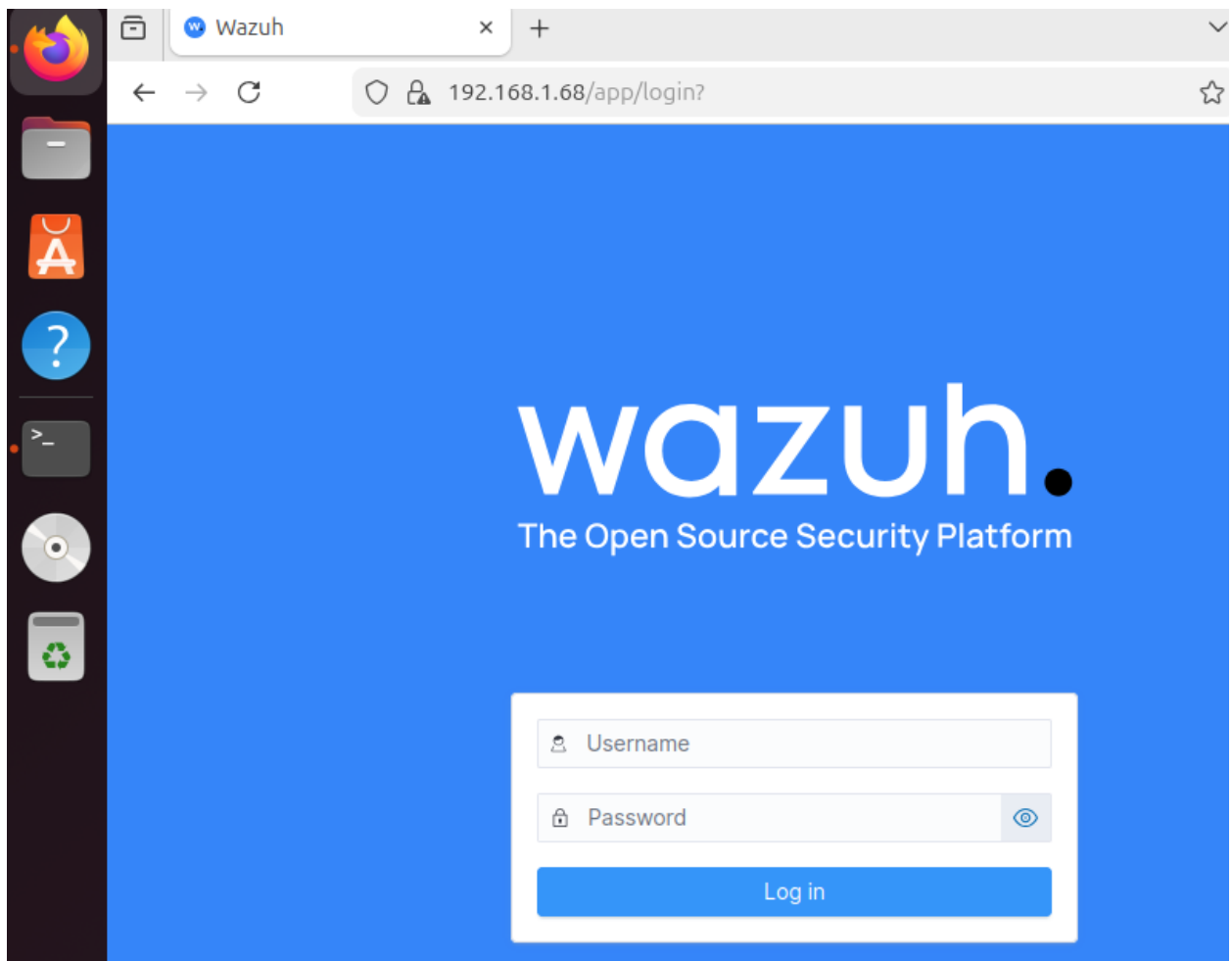
User: admin

Password: Fm8D.sbQPZXiNtXhEe?aYatKMow+cXD7

3. Accessing the Wazuh Dashboard

Once installation was complete, I brought back the Ubuntu VM's IP address:

192.168.1.68 and pasted it in Firefox.



← → ↻ 192.168.1.68/app/server-apis#/settings?tab=api ☆ ⓘ ⌵

W. Server APIs a ⓘ

API Connections

⊕ Add API connection ↻ Refresh ↻ Check updates ⓘ ☐ Disable updates notifications ⓘ

From here you can manage and configure the API entries. You can also check their connection and status.

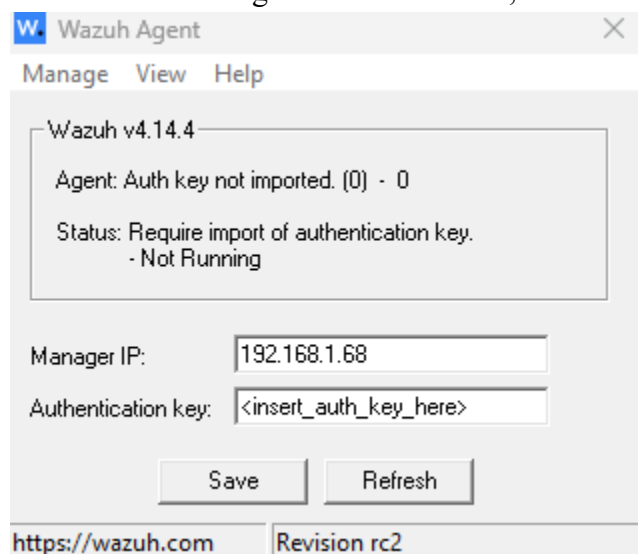
🔍 Search...

ID	Cluster	Manager	Host	Port	Username	Status	Version	Updates status	Run as	Actions
default	Disable	Ubuntuv	https://1	550	wazuh-wui	● Online	v4.12.0	● Available updates	👁	- ★ ↻

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4. Installing the Wazuh Agent (Windows Host)

1. Downloaded the latest Wazuh Agent MSI installer from the official Wazuh documentation page for Windows.
<https://documentation.wazuh.com/current/installation-guide/wazuh-agent/wazuh-agent-package-windows.html>
2. Ran the MSI installer using default settings.
3. When the installation was completed, I opened the Wazuh Agent Manager from the Start Menu.
4. In the Manager IP Address field, I entered the IP address of the Ubuntu VM.



5. Registering the Agent with the Manager

5.1 Generate the Agent Key on the Ubuntu Manager

On the Ubuntu VM, run the agent management utility

```
sudo /var/ossec/bin/manage_agents
```

Follow the interactive menu:

- Select A to add a new agent
- Assign a name, WAgent
- Enter the Windows machine's IP address - 192.168.56.1
- Confirm to create, it will be assigned an ID
- Select E to extract the key, then enter the agent ID
- Copy the full key string that is displayed

```

* Wazuh v4.12.0 Agent manager.          *
* The following options are available: *
*****
(A)dd an agent (A).
(E)xtract key for an agent (E).
(L)ist already added agents (L).
(R)emove an agent (R).
(Q)uit.
Choose your action: A,E,L,R or Q: A

- Adding a new agent (use '\q' to return to the main menu).
Please provide the following:
  * A name for the new agent: WAgent
  * The IP Address of the new agent: 192.168.56.1
Confirm adding it?(y/n): y
Agent added with ID 001.

*****
* Wazuh v4.12.0 Agent manager.          *
* The following options are available: *
*****
(A)dd an agent (A).
(E)xtract key for an agent (E).
(L)ist already added agents (L).
(R)emove an agent (R).
(Q)uit.
Choose your action: A,E,L,R or Q: E

Available agents:
  ID: 001, Name: WAgent, IP: 192.168.56.1
Provide the ID of the agent to extract the key (or '\q' to quit): 001

Agent key information for '001' is:
MDAxIFdBZ2VudCAxOTIuMTY4LjU2LjEgYmU4OWQ2ZWZkMWU0MDAyY2Y5OWViMGM4NDJjNmJjZTIwMjY5ZjYyOTE
4YzhZjliMjdjYTY5YTEwMmRlNWFnNQ==

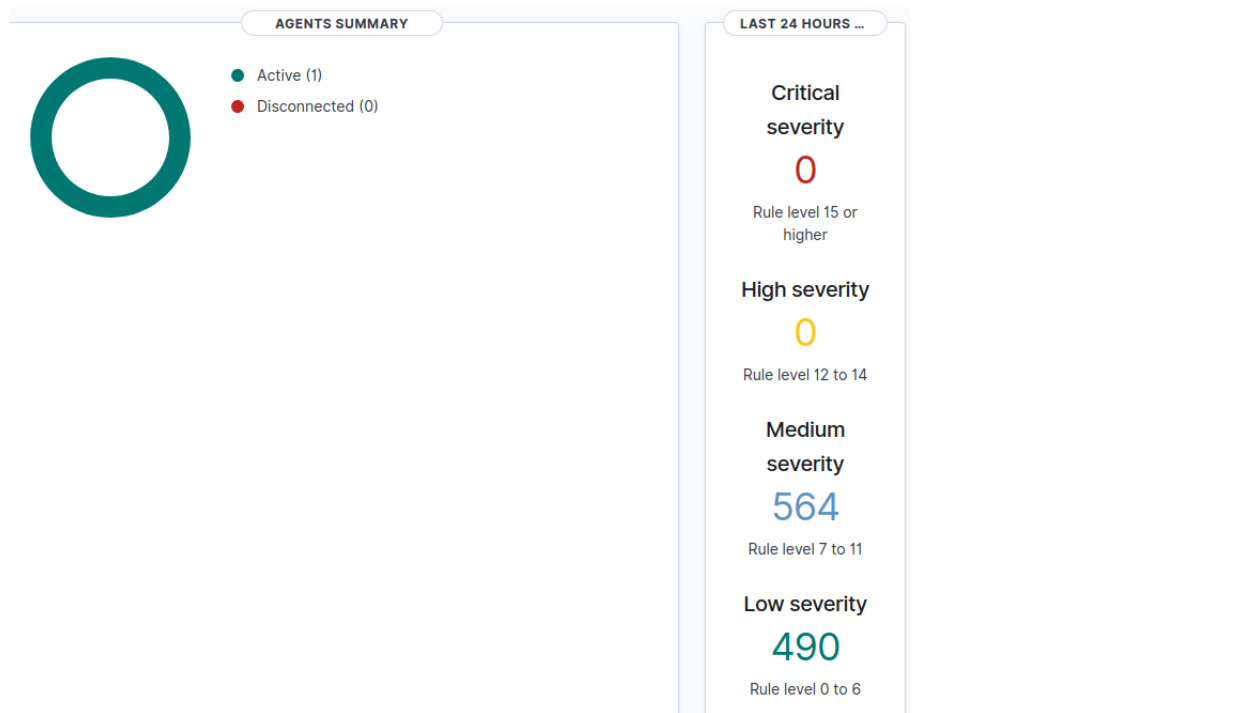
```

MDA1IFdBZ2VudCAxOTIuMTY4LjU2LjEgYmU4OWQ2ZWZkMWU0MDAyY2Y5OWViMGM4NDJjNmJjZTIwMjY5ZjYyOTE4YzhZjliMjdjYTY5YTEwMmRlNWFnNQ==

5.2 Apply the Key on the Windows Agent

5. Open the Wazuh Agent Manager GUI on Windows.
6. Paste the copied key into the Authentication Key field.
7. Save and apply the key.
8. Go to Manage > Restart to restart the agent service.

Go back to the Wazuh dashboard. The Agent Summary should now show one active agent. Click it to confirm your Windows machine is listed with its OS and IP details.



Agents (1)

[Deploy new agent](#) [Refresh](#) [Export formatted](#) [More](#) [Settings](#)

☒ Show only outdated

status=active									WQL
☐	ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
☐	005	WAgent	192.168.1.212	default	Microsoft Windows 11 Home 10.0.26200.8039	node01	v4.12.0	● ⓘ	👁️ ⋮

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6. File Integrity Monitoring (FIM)

Wazuh's Syscheck module monitors a directory in real time. Any file created, modified, or deleted inside the folder sends an alert to the Wazuh Manager dashboard.

6.1 Creating a Test Folder on Windows

Create a new folder on the Windows machine to use as the monitored directory:

C:\Users\Matth\Wazuh

6.2 Editing the Agent Configuration File

Open Notepad as Administrator, then open the following file:

C:\Program Files (x86)\ossec-agent\ossec.conf

Find the File Integrity Monitoring (syscheck) section. Inside the directories block, add a new line pointing to your test folder:

```
<directories realtime="yes">C:\Users\YourName\WazuhTest</directories>
```

Save the file with Ctrl+S.

```
<directories realtime="yes">%PROGRAMDATA%\Microsoft\Windows\Start Menu\Programs\Startup</directories>  
<directories realtime="yes">C:\Users\Matth\Wazuh</directories>
```

6.3 Restart the Agent

Restart the Wazuh agent service to apply the configuration change: Manage > Restart in the Wazuh Agent Manager GUI.

7. Verifying the Setup

7.1 Open the Wazuh Dashboard and go to the Windows agent's Integrity Monitoring section.

7.2 On Windows, create a new text file inside the monitored folder

7.3 Refresh the dashboard, a file creation alert appeared with the filename and agent details.

7.4 Delete the file, then refresh again, a file deletion alert appeared.

▼	syscheck.path	▼	syscheck.event	▼	rule.des...	▼	rule.level	▼	rule.id	▼
	c:\users\matth\wazuh\test.txt		deleted		File deleted.		7		553	
	c:\users\matth\wazuh\test.txt		added		File added ...		5		554	
	c:\users\matth\wazuh\new text document.txt		deleted		File deleted.		7		553	
	c:\users\matth\wazuh\new text document.txt		added		File added ...		5		554	

8. Summary

This lab demonstrated a complete Wazuh SIEM deployment from scratch, covering potential SOC work. The key points were:

- Installed and configured the Wazuh Manager on an Ubuntu VirtualBox VM
- Accessed and explored the Wazuh web dashboard
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